



Faculty of Engineering

Department of Biomedical Engineering

Non-Contact Type Clinical Thermometer

Housing Assembly

PART NUMBER:01-MIY-010

REV:1

DATE:10.05.2026



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FIRST RELEASE

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REVISIONS

Doc. No: 01-MIY-010

Non - Contact Thermometer

Rev1

Page 2/6

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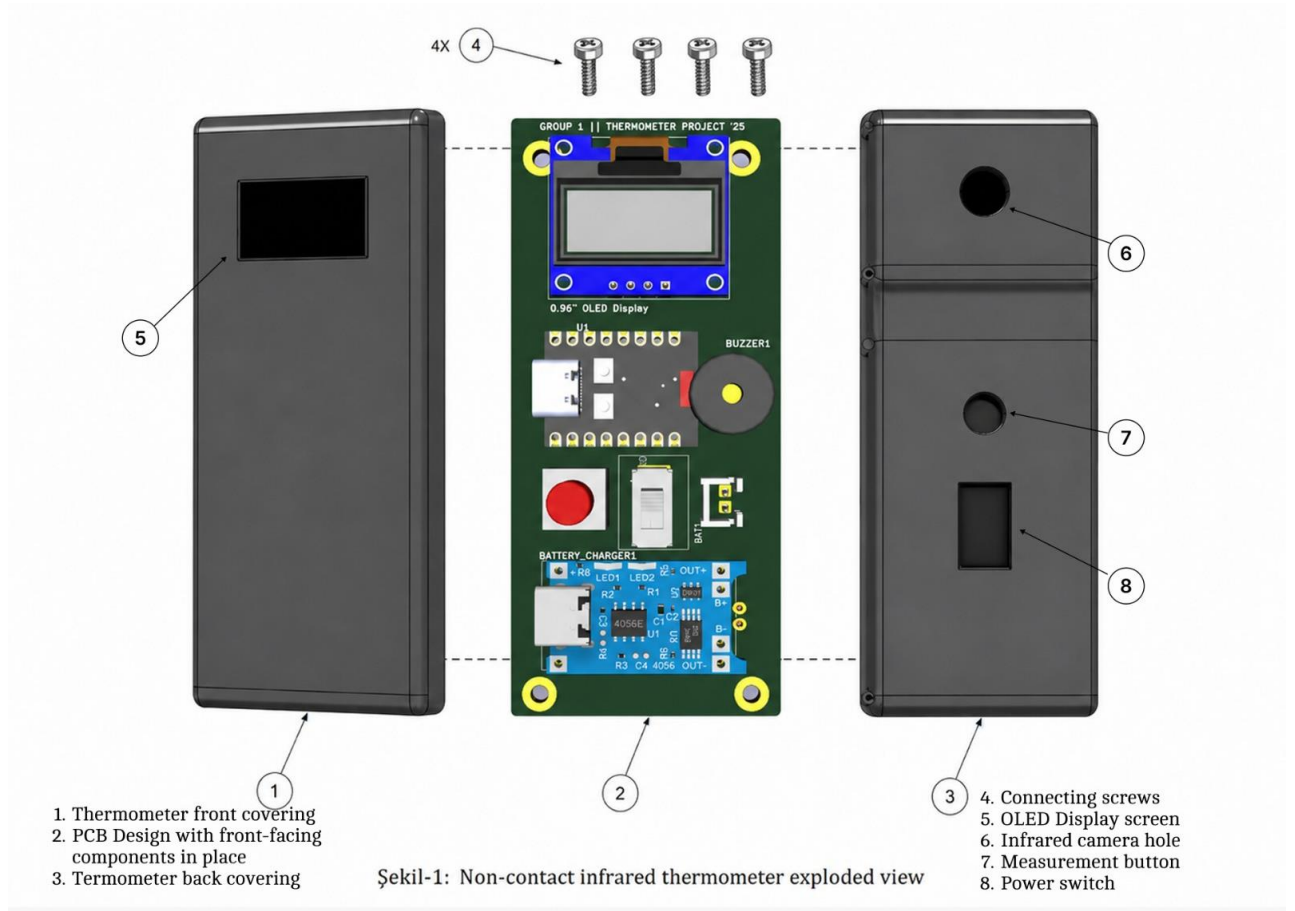
TABLE OF CONTENTS

CHANGE RECORDS.....	2
FIRST RELEASE.....	2
REVISIONS.....	2
1.DRAWINGS.....	4
2.PART LIST (BILL OF MATERIAL).....	5
3.TECHNICAL DESCRIPTIONS.....	6

Doc. No: 01-MIY-010	Non - Contact Thermometer	Rev1	Page 3/6
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1. DRAWINGS

1.1 COMPONENTS LOCATIONS & ELECTRICAL INTERFACE



1.3 BLOCK DIAGRAM

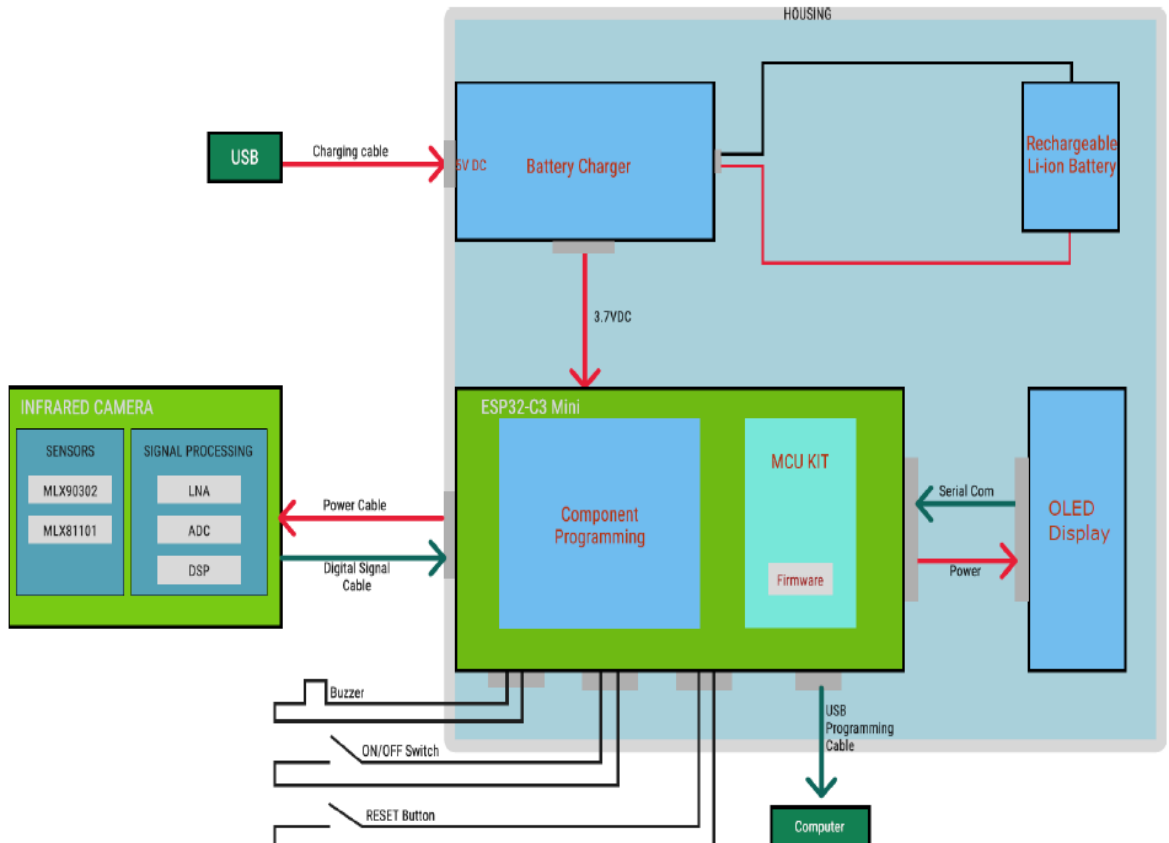


Figure 2:Block Diagram of The Design



2. PART LIST (BILL OF MATERIAL-BOM)

Item	P/N	Description	Location	Quantity
1	01-PRT-001	ESP32-C3 Mini Control Board Assembly		1
2	01-PRT-002	Infrared Camera Module Assembly		1
3	01-PRT-004	OLED Display Module		1
4	01-PRT-003	Rechargeable Li-ion Battery		1
5	01-PRT-005	ON/OFF Push Button	S1	1
6	01-PRT-012	Housing, Bottom		1
7	01-PRT-012	Housing, Top		1
8	01-PRT-007	Buzzer	BZ1	1
9		Screw, M3 × 12 mm	H1-H4	4

3. TECHNICAL DESCRIPTIONS

3.1. The non-contact infrared thermometer is comprised of three main sub-assemblies: the Main Control Board Assembly, Infrared Sensor Assembly, and Power Supply Assembly.

3.2. The Infrared Sensor Assembly measures body temperature without physical contact by detecting the infrared radiation emitted from the target surface. The sensed infrared energy is converted into an electrical signal by the infrared temperature sensor. The sensor output is transmitted to the Main Control Board through the corresponding signal connections for further processing.

3.3. The Main Control Board Assembly provides power distribution to all electronic circuits, processes the sensor data received from the infrared sensor, converts the analog signals into digital values through the microcontroller, and displays the calculated temperature measurement on the OLED display screen. The board also controls the buzzer, measurement button, and power switch functions.

3.4. During assembly, the Main Control Board is first positioned into the bottom housing enclosure. The OLED display, infrared sensor alignment, measurement button, and power switch are then checked and secured in their designated locations before final enclosure assembly.

Doc. No: 01-MIY-010	Non - Contact Thermometer	Rev1	Page 6/6
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3.5. Before fastening the top housing cover, an operational verification test is performed to confirm proper device functionality, including display operation, sensor response, and power control. After successful verification, the enclosure is secured using four screws and properly tightened.

3.6. The completed device will undergo functional and performance testing in accordance with the specified Acceptance Test Procedure to verify accurate temperature measurement and overall system operation.

Doc. No: 01-MIY-010	Non - Contact Thermometer	Rev1	Page 7/6
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